

DAY 5 — INTEGRATED CAPSTONE

Group Scenario Briefs

Cloud, Database & Cybersecurity Essentials — Integrated IT Fundamentals Training

Five groups, five distinct scenarios. Each group works only from its own page — don't compare answers across groups until the readout. Every scenario pulls in something from all five days: a compute and region decision, a database choice, and one security control you have to name and defend.

Time: ~45 minutes to work through your scenario, 5–7 minutes to present.

Format: short written answers to the five task items, plus a simple architecture sketch and a one-line final recommendation.

GROUP 1

Crude Oil Production & Royalty Reporting Portal

Licensed upstream operators (oil & gas companies) submit monthly crude oil production and royalty data for regulator review, validation, and audit. Field officers in Lagos, Port Harcourt, Warri and Kaduna also log inspection visit records against the same data.

BUSINESS REQUIREMENTS

#	Requirement	Detail
1	Users	~150 external operator company users plus ~40 internal regulator staff. Usage is heavy in the first 10 business days of each month, spiky the rest of the time.
2	Availability	Must be available during business hours nationwide. Brief overnight outages are tolerable, but not during the monthly submission window.
3	Data sensitivity	Production volumes and royalty figures are commercially sensitive and feed directly into government revenue calculations.
4	Data residency	Regulator policy prefers data to stay within Africa. If no suitable Azure region exists, the choice must be documented and justified on compliance grounds.
5	Team skills	A small 4-person IT team, strong in Windows Server and SQL, with limited Linux/Kubernetes experience and no dedicated 24/7 operations team.
6	Budget	Constrained public-sector budget. Predictable, minimal-maintenance costs are preferred over raw performance headroom.
7	Growth	Expected to expand to cover downstream fuel distribution monitoring within 18 months, roughly doubling data volume and adding ~100 more external users.
8	Integration	Must eventually integrate with an existing on-prem licensing database (a future phase, not this one).
9	Audit	Every data change must be traceable to a specific user and timestamp — a regulatory audit trail requirement.

YOUR TASK

- 1. Compute** — IaaS or PaaS? One sentence tying your choice to the team's skill level and how spiky or steady your usage pattern is.
- 2. Database** — Azure SQL Database, Cosmos DB, or something else? One sentence on why, tied to whether your data is structured/relational or more flexible/high-volume.
- 3. Region** — Pick a region and justify data residency in one sentence. There is no Azure region in Nigeria — state how you'd justify your choice anyway.
- 4. Security control** — Name one Day 4 control you'd apply first (e.g. Conditional Access, sensitivity labels, PIM, Defender for Cloud) and why it matters most for this scenario.
- 5. Architecture sketch** — A simple box-and-arrow diagram showing compute, database, and where your chosen security control sits. It doesn't need to be polished.

FINAL RECOMMENDATION (fill in the blanks)

For [your scenario], we recommend [compute] + [database] in [region], with [security control] as our first line of defense, because _____.

Pipeline & Facility Sensor Monitoring

Pressure, flow, and temperature sensors across pipeline stations in Port Harcourt, Warri, and Kaduna stream readings every few seconds. Engineers need near-real-time dashboards and historical trend analysis to catch equipment failures early.

BUSINESS REQUIREMENTS

#	Requirement	Detail
1	Users	~25 internal engineering and operations staff monitoring dashboards. No external users.
2	Availability	Near-24/7. Pipeline safety monitoring cannot tolerate extended outages, including overnight.
3	Data sensitivity	Operational telemetry (pressure, flow, temperature). Not personally identifiable, but commercially sensitive with respect to throughput volumes.
4	Data residency	Same regulator policy applies: prefer an Africa-based region, and document a compliance justification if none is available.
5	Team skills	The same small 4-person IT team, with some exposure to IoT/telemetry ingestion but no dedicated data-engineering staff.
6	Budget	Constrained overall, but the regulator is willing to spend more on this system than the others given its safety criticality.
7	Growth	Sensor count is expected to double as more pipeline segments and pumping stations are instrumented.
8	Integration	Needs to eventually feed alerts into a control-room, SCADA-adjacent alerting system (a future phase).
9	Audit	Sensor readings and any manual overrides must be timestamped and traceable for incident investigations.

YOUR TASK

- 1. Compute** — IaaS or PaaS? One sentence tying your choice to the team's skill level and how spiky or steady your usage pattern is.
- 2. Database** — Azure SQL Database, Cosmos DB, or something else? One sentence on why, tied to whether your data is structured/relational or more flexible/high-volume.
- 3. Region** — Pick a region and justify data residency in one sentence. There is no Azure region in Nigeria — state how you'd justify your choice anyway.
- 4. Security control** — Name one Day 4 control you'd apply first (e.g. Conditional Access, sensitivity labels, PIM, Defender for Cloud) and why it matters most for this scenario.
- 5. Architecture sketch** — A simple box-and-arrow diagram showing compute, database, and where your chosen security control sits. It doesn't need to be polished.

FINAL RECOMMENDATION (fill in the blanks)

For [your scenario], we recommend [compute] + [database] in [region], with [security control] as our first line of defense, because _____.

PPE & Safety Stock Inventory Management

An organization-wide extension of the inventory system built earlier this week — now covering every regulator facility, with low-stock alerts and supplier reorder workflows.

BUSINESS REQUIREMENTS

#	Requirement	Detail
1	Users	~30 internal facility managers and store officers across all regulator facilities. No external users.
2	Availability	Needed during normal working hours only. A few hours of overnight downtime is acceptable.
3	Data sensitivity	Stock levels and supplier details. Low sensitivity — no personal or regulated financial data involved.
4	Data residency	Same regulator policy applies. This is the lowest-stakes system for residency — worth deciding as a group whether it deserves the same weight as Group 1's.
5	Team skills	The same 4-person IT team. This system is the closest to what they already manage day-to-day.
6	Budget	The most budget-constrained of all five systems — treated as a lower-priority internal tool.
7	Growth	Expected to scale from 4 facilities today to all ~15 regulator facilities within a year.
8	Integration	Should eventually integrate with the procurement/finance system for automatic reordering.
9	Audit	Stock movements must be traceable to a specific staff member — a loss-prevention requirement.

YOUR TASK

- 1. Compute** — IaaS or PaaS? One sentence tying your choice to the team's skill level and how spiky or steady your usage pattern is.
- 2. Database** — Azure SQL Database, Cosmos DB, or something else? One sentence on why, tied to whether your data is structured/relational or more flexible/high-volume.
- 3. Region** — Pick a region and justify data residency in one sentence. There is no Azure region in Nigeria — state how you'd justify your choice anyway.
- 4. Security control** — Name one Day 4 control you'd apply first (e.g. Conditional Access, sensitivity labels, PIM, Defender for Cloud) and why it matters most for this scenario.
- 5. Architecture sketch** — A simple box-and-arrow diagram showing compute, database, and where your chosen security control sits. It doesn't need to be polished.

FINAL RECOMMENDATION (fill in the blanks)

For [your scenario], we recommend [compute] + [database] in [region], with [security control] as our first line of defense, because _____.

Oil Spill & Environmental Incident Reporting

Field officers and the general public can both report spills or environmental incidents, with photo upload and GPS location, alongside an internal investigation and review workflow.

BUSINESS REQUIREMENTS

#	Requirement	Detail
1	Users	The general public (unlimited, unauthenticated submissions) plus ~20 internal reviewers and investigators.
2	Availability	True 24/7. Incidents and spills don't wait for business hours — downtime directly risks missed reports.
3	Data sensitivity	High. Some reports involve active investigations, potential litigation, and personal data (reporter contact details) under NDPA/NDPR.
4	Data residency	Same regulator policy applies. The public-facing nature of this system raises the compliance stakes further.
5	Team skills	The same small IT team — this is the most operationally critical system they will be asked to run.
6	Budget	More flexible than the other systems, given the regulatory and legal exposure of getting this one wrong.
7	Growth	Reporting volume is expected to grow as public awareness campaigns expand.
8	Integration	Needs to eventually integrate with a GIS/mapping system to plot incident locations.
9	Audit	Every report, status change, and investigator action must be fully traceable — this data may become legal evidence.

YOUR TASK

- 1. Compute** — IaaS or PaaS? One sentence tying your choice to the team's skill level and how spiky or steady your usage pattern is.
- 2. Database** — Azure SQL Database, Cosmos DB, or something else? One sentence on why, tied to whether your data is structured/relational or more flexible/high-volume.
- 3. Region** — Pick a region and justify data residency in one sentence. There is no Azure region in Nigeria — state how you'd justify your choice anyway.
- 4. Security control** — Name one Day 4 control you'd apply first (e.g. Conditional Access, sensitivity labels, PIM, Defender for Cloud) and why it matters most for this scenario.
- 5. Architecture sketch** — A simple box-and-arrow diagram showing compute, database, and where your chosen security control sits. It doesn't need to be polished.

FINAL RECOMMENDATION (fill in the blanks)

For [your scenario], we recommend [compute] + [database] in [region], with [security control] as our first line of defense, because _____.

Contractor & Staff Identity, Access & Compliance Training

Tracks which contractors and staff have completed mandatory safety and security training, and manages who can access which facility systems — including the other four systems in this capstone.

BUSINESS REQUIREMENTS

#	Requirement	Detail
1	Users	~40 internal staff plus ~200 contractors across all facilities. This system controls who can access every other system in the capstone.
2	Availability	Needed during business hours. Brief downtime is tolerable, but it blocks access to everything else while it's down.
3	Data sensitivity	High. Personal data — identity verification, training records — falls directly under NDPA/NDPR.
4	Data residency	Same regulator policy applies. The personal-data element adds extra weight to the residency/compliance justification for this system specifically.
5	Team skills	The same 4-person IT team. This system requires the deepest identity and security expertise of any group's system.
6	Budget	Constrained, but treated as foundational infrastructure the other four systems depend on.
7	Growth	Contractor headcount is expected to grow significantly as field operations expand.
8	Integration	Must eventually integrate with the HR system for automatic onboarding and offboarding.
9	Audit	Every access grant, revocation, and training completion must be traceable — this is the audit trail for the audit trail.

YOUR TASK

- 1. Compute** — IaaS or PaaS? One sentence tying your choice to the team's skill level and how spiky or steady your usage pattern is.
- 2. Database** — Azure SQL Database, Cosmos DB, or something else? One sentence on why, tied to whether your data is structured/relational or more flexible/high-volume.
- 3. Region** — Pick a region and justify data residency in one sentence. There is no Azure region in Nigeria — state how you'd justify your choice anyway.
- 4. Security control** — Name one Day 4 control you'd apply first (e.g. Conditional Access, sensitivity labels, PIM, Defender for Cloud) and why it matters most for this scenario.
- 5. Architecture sketch** — A simple box-and-arrow diagram showing compute, database, and where your chosen security control sits. It doesn't need to be polished.

FINAL RECOMMENDATION (fill in the blanks)

For [your scenario], we recommend [compute] + [database] in [region], with [security control] as our first line of defense, because _____.

Instructor Reference — Presentation Rubric

Use the same rubric across all five groups. Read this page after group work begins, not before — it isn't part of the student handout.

Criterion	What to look for
Reasoning, not memorization	Can the group explain *why*, tied to their scenario's specific requirements — not just name a service.
Cross-day integration	Did they pull in something from cloud, data, and security — not just one day's material.
Residency reasoning	Did they engage with the "no Nigerian region" problem rather than skip it.
Presentation clarity	Is the sketch and explanation understandable to someone who didn't build it.

Common things to probe in discussion

- A group picks IaaS "for control" without weighing the team-skills/maintenance requirement.
- A group picks a region purely on lowest listed price without addressing residency/compliance reasoning.
- A group forgets the audit-trail requirement when comparing database services — this should favor managed services with built-in change tracking and backup history over self-managed alternatives.
- Group 5 in particular should recognize that its system is the security dependency for the other four — push them on what happens to Groups 1-4 if their identity layer is weak.